AGI ALPHA FLYWHEEL

The Paid-Progress Option — Executive Brief

Vincent Boucher — President of MONTREAL.AI 27 October 2026

A useful milestone has been delivered; profitability remains conditional

The **Reuse Investment Gate Pack** is an executable digital offering for an industrial plant owner, reuse developer or adviser deciding whether to fund the next water-reuse study. It supplies source-linked inputs, corrected cash economics, conditional quality checks, explicit stop/defer reasons and a next-measurement plan. A negative opportunity can be a useful accepted deliverable if that question and scope were agreed in advance.

The delivered result

A published process-water case states less than one year's payback. Its own cost table instead implies **3.37 years**, before omitted costs: EUR82,000 stated annual savings less EUR28,630 cash operating expense leaves EUR53,370 against EUR180,000 capital. At an assumed 12% discount rate over ten years, the partial-input NPV is EUR121,552.

The pack returns **DEFER**. Discharge limits are not a buyer's process-water specification, and long-run performance, complete costs and six other evidence domains are unresolved. This is an actionable digital screening result, not a new treatment technology, physical validation or plant authorization. Source: Marjana Simonič, *Membranes* 11(12), 976 (2021), DOI 10.3390/membranes11120976.

Commercial scope and modeled economics

The selected offer is **US$3,500**, against a transparent equivalent-scope human estimate of US$5,200. The modeled contract includes a reserved 30% deposit, acceptance of the agreed work, one bounded correction and refund on failed final acceptance. No customer was contacted and no contract or payment occurred.

Complete-cost scenario	Mean project profit	Profitable histories
Conservative	**−US$9,000**	0 of 256
Central	**+US$2,962**	152 of 256
Favorable	**+US$77,279**	253 of 256

The model covers 26 selling weeks plus seven settlement weeks and requires **US$10,000 external seed**. It includes unsuccessful work, correction, acquisition, support, bad debt, setup and reserved obligations. These primary figures are an explicit accounting-correction replay: initial frozen runs omitted settlement overhead. Their original central result was US$6,477; reserving the added expense also restricts intake, reducing the corrected mean to US$2,962. The price and original seeds were not retuned.

The campaign is not robust across demand assumptions. The conservative price attracts zero orders and acquisition consumes the risk budget. The central case loses money in 40.6% of modeled histories. Buyer-price interpretation, acquisition, acceptance and residual effort can reverse the result. Actual collections are **US$0**; complete actual cost and profit are **unknown**.

What comes next—and what the evidence does not justify

Three separate accounts

Provider profit comes from the specified service fee less its full costs. Customer value comes from usable output or an improved decision, net of price and adoption burden. Future plant or venture upside belongs to whoever later owns the relevant rights and bears its risks. These are not additive earnings.

A separate **hypothetical** empirical test has EUR30,621 gross information value and EUR22,348 net value under a 0.50 viability prior, 0.80 sensitivity/specificity, US$8,000 test fee, fixed scenario conversion of 1.10 USD/EUR and EUR1,000 delay burden. It changes a modeled stop decision after a positive signal. Those probabilities are uncalibrated planning assumptions. This value is not attributed to the delivered desk pack, and the physical project remains deferred.

Finance and selective continuation

The corrected central first-job funding threshold is **US$8,945**, or **US$9,295 under the shared shock**, including committed overhead, reserves and one acquired lead that books. It does not guarantee a first sale. Zero-seed operation refuses.

All optional US$1,200/US$2,500 reusable-module investments remain **STOP** after construction, maintenance, failure, scope fit and lost delivery capacity. Available money alone is insufficient. The study demonstrates neither retained Alpha nor compounding.

Technical evidence and recovery

Ordinary and expert implementations each satisfy 62 of 64 original synthetic confirmation cases. One discrepancy violates an exact negative-NPV rule; another reveals decimal/binary interpretation ambiguity. The conventional evaluator matches the declared cases but also supplies the numerical oracle; its self-match is not independent validation. The final wrapper uses it, preserves both failures and passes 14 post-confirmation diagnostics with separate exact-arithmetic checks. No expert-model advantage or improved fresh-confirmation rate is claimed.

The corrected economics ledger contains 185,135 event rows across 768 histories. An independent Decimal implementation checks all cash, accrual and reserve identities. Local native accounting keeps unknown actual costs unknown; existing FusionPlan verification is exercised. ENS is unverified; MARK is not executed; no live token or payment transaction occurs.

Scale is a defined scenario

6,000 qualified projects at the historical EUR180,000 case capital would represent **EUR1.08 billion cumulative plant deployment by 2040**. Eligibility, current costs and adoption are unverified. This is neither provider revenue nor valuation. At US$3,500 per pack, the same 6,000 orders would produce US$21 million gross service receipts before costs and competition.

Decision

The completed package supports a **separately authorized, bounded buyer pilot** of the screening service, with measured input effort, acceptance, corrections, bills and collection. It does not justify automatically funding the full modeled campaign, buying the hypothetical field experiment or committing plant capital. Useful progress can precede the larger answer; continuation must earn its own evidence and funding.

The research package includes the full paper, source, executable study, frozen inputs, original and corrected ledgers, readable delivery pack, failure records and an unposted LinkedIn draft. Distribution is a pinned review edition; canonical publication follows existing-owner qualification.